
Software Requirements Specification

for

VHEcoPoint: A Smart Waste Segregation Station with QR-Based Incentive System for Victorian Heights Subdivision

Version 1.0

Prepared by

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Project Overview

VHEcoPoint is a hardware-integrated smart waste segregation and rewards system developed for deployment at Victorian Heights Subdivision. The system is composed of physical VHEcoPoint Station as a base and a user interface connected directly to “VictorianPass” website. An Internet of Things (IoT) based waste bin that can identify and segregate waste based on material category of; paper or cardboard, plastic (PET bottles) and metal (aluminum cans). That aims to promote community engagement through incentive points based on weight of the deposited materials that the users can utilize for amenity free 1 hour discount. The system provide value as it provides opportunity for a cycle of efficient waste management and user engagement by way of providing engagement and awareness towards proper waste disposal preventing non-biodegradable and biodegradable waste from mixing as prolonged material breakdown that can cause floods, landfill and health risk.

Victorian Heights Subdivision, a 2,220-household residential community located at Dahlia, Fairview, Brgy. Sauyo, Quezon City. The current situation regarding proper segregation of waste, as residents often neglect reduction of waste build-up and recycle opportunity. Despite the regular waste collection services, according to most of the residents, there are reports that majority of households mainly store water bottles for daily in a weekly basis. furthermore, most households in the subdivision possess old scrap materials, including obsolete documents, discarded small to medium Cardboard boxes, and leftover canned goods and soda cans that are made of aluminum. The community lacks structured system to encourage proper segregation and promote resident and subdivision cleanliness and well-being. Which causes waste collection difficulty as large quantity of these materials cannot be collected in bulk and prevent other waste that needs to be collected that day. As Victorian Heights Subdivision does not have regulation for separation of waste from the source, as the work was given to waste collection for manually separation of general waste on the spot.

Republic Act No. 9003: Mandatory Segregation of Solid Wastes (2001, Section 21), states that segregation of waste should primarily be conducted at the source. In this republic act, addresses that There shall be a separate container type of waste for all sources and solid waste containers should have label identifying them as “compostable”, “non-recyclable”, “recyclable” or any classification for different types. This practice was left out, there are no execution to spread awareness regarding this ordinance, which makes resident do not follow any waste management practices. At the same time, amenities such as the Multi-Purpose Building, courts, and clubhouse often experience scattered materials after usage whereby the staff often left with additional work that they need to fix before another batch of customers use the said amenity.

VHEcoPoint is designed to solve these problems as an engaging alternative for traditional wastebins. The system key function includes real-time material differentiation such as paper/cardboard, PET plastic bottles and metal (soda can and tin can) using sensors. Identify users with QR-based identification, weight-based incentive pointing system for amenity discount. With this the physical layer of the system will send the data to the Victorian Pass' VHEcoPoint user interface, to which users can access and overview their activity per session. The system enables users to have a centralized interface where they can securely navigate amenities while accumulating incentives points for 1 free hour discounts.

Functionality Requirements

Table 1: Functionality Requirements

ID	Requirement Name	Description (The system shall..)	Priority
FR-01	User Registration	The system shall allow users (Resident, Admin) to create accounts, while providing required credentials.	High
FR-02	Resident login	The system shall allow user to login to their respective registered accounts.	High
FR-03	User QR Download / Acquisition	The system shall allow Residents to download their QR code as their unique ID to access VHEcoPoint Physical and VictorianPass' VHEcoPoint web interface.	High
FR-03	Verification and Session Creation	The Raspberry Pi 4b shall connect to internet connection using HTTPS and access VictorianPass's domain/ IP address to create session, after the physical hardware (GM65 QR / Barcode scanner) scanned the users registered account QR code.	High
FR-04	Raspberry Pi QR Verification	The Raspberry Pi shall check the users QR code.	High
FR-05	Raspberry Pi Notified	The system shall notify the raspberry Pi that account was verified.	High
FR-06	VHEcoPoint Account Access Notification	The VictorianPass Website shall notify the user that the account was successfully connected to the VHEcoPoint.	High
FR-07	Physical Layer interface	The Raspberry Pi shall display the user identification and incentive points on a LCD 1602 display module in VHEcoPoint.	Medium
FR-08	Waste Deposit identification	The VHEcoPoint shall be able to differentiate material such as paper and plastic using ESP32-CAM while streaming YOLOv11 that is trained from Raspberry PI, while inductive sensor for reading metal. The readings will be processed to identify the object detected	High

FR-09	Weight-Based Incentive	The VHEcoPoint shall be able to weigh the object detected, while giving reading to the system that will be processed as data for incentive using HX711 loadcell module: • Plastic (PET bottles $\leq 1000\text{ml}$): 55 points per kg • Aluminum cans: 140 points per kg • Paper & Cardboard (combined): 30 points per kg	High
FR-10	Data processing	The system shall process the data from the sensors and send it to the physical interface and the web interface of VHEcoPoint.	High
FR-11	Object Segregation	The system shall read and move the servos accordingly to make way for materials to go to its appropriate bin (plastic, paper / cardboard and metal).	High
FR-12	Physical to Web Transmission	The Raspberry Pi shall send the readings to the VictorianPass' VHEcoPoint module	High
FR-13	Incentive Display	The Data (Incentive points) shall reflect to VHEcoPoint user interface	High
FR-14	User Activity Overview	The system shall allow overview of user VHEcoPoint activity	Medium
FR-15	Activity Point Gathering	The system shall allow accumulation of incentive points	High
FR-16	Discount Selection	The system shall allow selection of amenity for discount	High
FR-17	Reserve Amenity	The system shall allow to reserve for the amenity.	Medium
FR-18	Amenity Downpayment	The system shows down-payment page, the user must upload their payment and reference code of the payment	High
FR-19	Reservation Finalization	The system must allow users to finalized their reservation	High
FR-20	Session timeout and exit	The system shall reset or end VHEcoPoint session if inactive or idle for 2 minute.	High
FR-21	Resident Logout	The system must allow Resident to logout of their account.	High
FR-22	Admin Login	The system allow admin to login with their registered account.	High
FR-23	Admin Dashboard	The system shall present the admin view of the VictorianPass Dashboard.	High
FR-24	Admin VHEcoPoint Dashboard	The system allows to have an overview of the overall operational and engagement of users on the system.	High
FR-25	Admin Logout	The system must allow admin to logout of their account.	High

Non-Functional Requirements

The Non-Functional Requirements for VHEcoPoint define how well the system operates to deliver a seamless, secure, and dependable experience for Victorian Heights residents, beginning with stringent performance benchmarks: hardware session-creation completes within 500 milliseconds, points are calculated in under 200 milliseconds per session, and both resident and admin VHEcoPoint dashboards render fully within three seconds under normal network conditions to ensure real-time responsiveness even as transaction volumes scale. In terms of usability and user experience, the station is architected for fully unsupervised self-service, requiring only periodic manual maintenance and bug checks while its visual design and responsive layout are faithfully aligned with the VictorianPass platform, allowing residents to view their point balances and recent transaction histories without navigating away from their primary interface. To uphold robust security and data integrity, all station API keys are stored using industry-standard password hashing, and the system strictly rejects requests with missing or invalid credentials before any business logic executes; furthermore, an immutable, complete event log captures every phase of every session from authentication and scanning to weighing, capping, and posting, ensuring that point transaction records can never be directly edited or deleted, thereby preserving an unalterable audit trail. On reliability, the system connects via a Local Area Network (LAN) within its deployment area, enforces all points-award processes as atomic, all-or-nothing transactions to eliminate duplicate credits even during retries or duplicate submissions, and guarantees at least 99% monthly uptime for all VHEcoPoint API endpoints (excluding scheduled maintenance), with interrupted sessions seamlessly recoverable through logged event histories. Compatibility is equally prioritized, as both the hardware bridge and web interface are rigorously tested to function across all standard desktop and mobile operating systems and browsers utilized by residents, guards, and administrators. Finally, to safeguard long-term operational resilience, the system automatically backs up all VHEcoPoint-related database tables as part of the platform's nightly routine, retaining both daily and weekly copies to ensure rapid data restoration when needed. Collectively, these

non-functional specifications ensure that VHEcoPoint not only functions correctly, but performs exceptionally well—delivering speed, security, reliability, and ease-of-use that truly build a greener, connected community.

References:

Republic Act 9003 (2001), “Ecological Solid Waste Management Act of 2000.”, Lawphil. [RA09003.p65](#)